

Lab 3 – Architectural Justification

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Architecture Selection

We chose Layered Architecture for the Corporate Mental Wellness & Counseling Platform System.

Style Comparison

Style	Pros	Cons / fit for this platform
Layered	Clear separation of concerns; simple to build, test and audit.	Tighter coupling and harder per-component scaling, which is acceptable at corporate-portal scale.
Microservices	Independent scaling, fault isolation and deployment.	Heavy operational cost and network latency; bookings and PII handling would span services, complicating consistency and privacy audits.
Client-Server	Centralised control and easy data consistency.	Single point of failure and bottleneck; gives no internal structure for isolating identity data.

Two Reasons for the Choice

- **Anonymity maps naturally onto layers.** The core requirement (FR-001) is that therapists never see corporate email or employee ID. In a layered design this rule lives in one place, the Authentication & Anonymization Service in the Business Layer. The Portal UI only receives a pseudo-ID and session token, and the Scheduler, Assessment Tracker and Notification Service address users by pseudo-ID only.
- **Well-bounded scope and transactional needs.** The system has two actors and a small set of workflows (assess: FR-002, book: FR-003, manage sessions: FR-004, track: FR-005) over one consistent data model. Slot booking must never double-book a therapist, which a single transactional data layer guarantees easily. Microservices would add distributed-data complexity with no matching benefit.

Security Advantage

Layers give strictly controlled entry points. The Portal UI can reach data only through the Business Layer interfaces (IAuthentication, IScheduling, IAssessment), and only the Business Layer talks to the Encrypted Data Store, over TLS and AES-256 encrypted at rest. The identity-to-pseudo-ID mapping sits behind a single restricted interface (IUserStore), so a compromised UI or therapist session cannot query employee PII. Authentication, authorisation and audit logging are enforced once at the Business Layer boundary rather than duplicated everywhere, directly supporting NFR-001 (no unauthorized access to employee identity or counseling data).

Performance Benefit

Calls between layers are in-process or on the same host, so there are no per-request network hops, giving low latency for interactive screens such as booking and dashboards. The Scheduler stores all times in UTC and converts to the user's local timezone only at the edge, so timezone handling (NFR-002) is computed once and consistently across global offices. Frequently read data such as therapist availability can be cached in the data-access interfaces, and the whole stack can be replicated behind a load balancer to handle peak load.